META SAMPLES COVERAGE ANALYSIS

Executive Summary:

This automated Python-based workflow identifies coverage gaps in high-Meta sample clusters using 1 km site buffers segmented into 60-degree azimuths, enabling targeted improvements via hygiene corrections or reorientations—without new CAPEX. As a Planning Lead focused on KPIs like churn and ARPU, this tool supports revenue optimization by prioritizing population-dense areas. Industry parallels show 15-33% reductions in churn or site counts through similar geospatial optimizations.

Expected Revenue & Churn Impact

Hygiene corrections and azimuth reorientations minimizes CAPEX leakage while boosting coverage in revenue-critical zones. With this model we are expected to:

- Identify sites for **physical optimization** which will result in opportunity to onboard/retain **uncovered Population.**
- This customer base will bring in **additional revenue**.
- Cumulative **churn** at these sites is expected to be reduced.

Methodology

1. Create a 1 km radius buffer around each site.
2. Divide each buffer into six 60-degree segments:

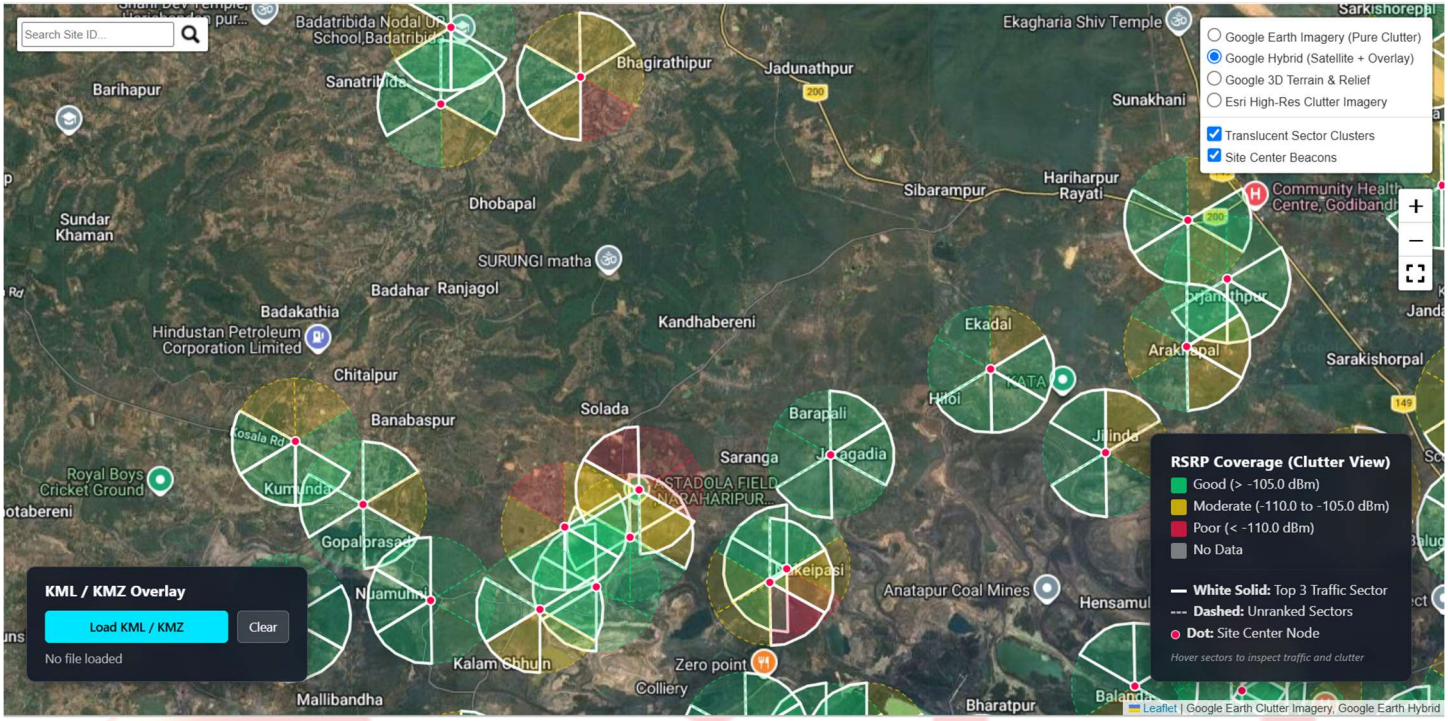
Segment	Degrees from N	Centre Degree
Segment_1	0-60	30
Segment_2	60-120	90
Segment_3	120-180	150
Segment_4	180-240	210
Segment_5	240-300	270
Segment_6	300-360	330

3. Map 4G Meta Samples to these segments across all sites.
4. Identify the top three segments by total Meta tiles.
5. Overlay GH RSRP sample tiles onto these segments.
6. Flag a top segment as "Uncovered" if average RSRP falls below -110 dBm.
7. Action Protocol:
   - If an existing sector points toward the uncovered segment (±30 degrees), prioritize hygiene corrections.
   - If no sector aligns within ±30 degrees and 4G traffic permits, reorient the nearest sector's azimuth.

Implementation

- Previously, manual analysis using MapInfo and QGIS proved time-intensive and non-reusable.
- This workflow has been fully automated using Python with libraries including Pandas, GeoPandas, Scikit-Learn, NumPy, and Shapely.
- Generative AI accelerated logic-to-code translation.
- Geospatial computations on large datasets presented performance challenges, resolved through iterative code optimization.
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Results



In above example we can see existing antenna azimuth probably looking at a different direction than GH samples indicating. We need hygiene check & correction here.

Way Forward

Model retuning and re-run of model across Odisha including Small Cells & ULS after we conclude in trial districts. Intelligent automations will help run large datasets without major modification and with shorter analysis lifecycle.

Thank You!
Team Odisha